

PROJECT REPORT

Data Structures & Algorithms Visualizer & Inventory Management System

1. Abstract

This project consists of two C++ applications developed as part of an internship task:

1. A Data Structures and Algorithms Visualizer
2. An Inventory Management System

The DSA Visualizer demonstrates sorting algorithms step-by-step using console-based visualization, while the Inventory System performs CRUD operations using file handling concepts.

The aim is to strengthen understanding of algorithms, data structures, and real-world programming logic.

2. Introduction

Data Structures and Algorithms form the foundation of computer science. Understanding how algorithms work internally is essential for problem solving and software development.

This project was designed to:

- Visualize sorting algorithms
 - Implement real-world inventory management
 - Improve modular programming skills in C++
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3. Objectives

- To visualize sorting algorithms step-by-step
- To implement multiple sorting techniques
- To build a functional inventory management system
- To understand file handling and data storage
- To improve problem-solving and coding structure

4. Technologies Used

- C++
 - Object-Oriented Programming
 - Arrays and Functions
 - File Handling (Inventory System)
 - Console-based interface
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5. Project Modules

5.1 Data Structures & Algorithms Visualizer

Features:

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Random Array Generator
- Step-by-step visualization using bars (*)
- Menu-driven interface

Working:

1. Generate random array
 2. User selects sorting algorithm
 3. Algorithm executes step-by-step
 4. Array is displayed after each operation
 5. Option to regenerate array
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5.2 Inventory Management System

Features:

- Add product
- Display products
- Search product
- Update product
- Delete product
- File-based storage

Working:

1. User selects operation
 2. Data is stored in file
 3. CRUD operations performed
 4. Updated data saved persistently
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6. System Design

DSA Visualizer:

- main.cpp → controls menu
- sorting.cpp → sorting logic
- utils.cpp → visualization + array generation

Inventory System:

- main.cpp → menu system
 - inventory.cpp → logic implementation
 - file storage → persistent data
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7. Algorithms Used

Bubble Sort

Repeated swapping of adjacent elements.

Selection Sort

Selection of minimum element repeatedly.

Insertion Sort

Builds sorted array gradually.

Merge Sort

Divide and conquer approach.

Quick Sort

Partition-based sorting algorithm.

8. Flow of Execution

1. Program starts
 2. Array is generated / inventory loaded
 3. Menu is displayed
 4. User selects operation
 5. Required function executes
 6. Output is displayed
 7. Program returns to menu
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9. Output Screenshots

(Insert screenshots here)

- Main menu
 - Sorting process
 - Final sorted array
 - Inventory system operations
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10. Learning Outcomes

- Understanding sorting algorithms deeply
- Practical implementation of DSA concepts
- File handling in C++
- Modular code design
- Real-world application development

11. Conclusion

This project successfully demonstrates both algorithm visualization and real-world system development. It improves understanding of DSA concepts and strengthens programming skills required in software development.

END OF REPORT